

Xiaohai Wang

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Security Analyst — SOC Tier 2 / Incident Response / Endpoint Security

Master's in Cybersecurity and Threat Intelligence with hands-on experience in SOC operations, endpoint threat investigation, and security awareness program development. Skilled in analyzing and responding to alerts from EDR platforms such as SentinelOne and CrowdStrike, producing comprehensive incident reports, and tuning detection rules to reduce false positives. Experienced in designing phishing simulations (phishing, vishing, smishing) and delivering security awareness reporting using platforms like KnowBe4 and Proofpoint. Strong background in malware analysis, incident handling, and cross-team collaboration for security product onboarding, tuning, and operational improvement.

EDUCATION

University of Guelph

Master of Cybersecurity and Threat Intelligence; GPA:92/100(A+)

Guelph, Canada

Sep 2024 – Sep 2025

Auckland University of Technology

Bachelor of Computer and Information Sciences (Exchange Program)

Auckland, New Zealand

Sep 2019 – Jun 2021

China Jiliang University

Bachelor of Engineering in Computer Science and Technology

Hangzhou, China

Sep 2017 – Jun 2021

Academic Excellence Scholarship (Three Times), Merit Student

SKILLS

SOC Operations & Incident Response: SentinelOne, CrowdStrike Falcon, triage, root cause analysis, containment, incident documentation, EDR alert investigation.

Threat Detection & Response: Rule tuning, playbook development, reducing false positives, SIEM log analysis, MITRE ATT&CK mapping.

Security Awareness & Human Risk Management: KnowBe4, Proofpoint, phishing/vishing/smishing campaigns, LMS management, awareness reporting.

Malware Analysis & Endpoint Security: Static/dynamic analysis, sandboxing, ransomware, keyloggers, persistence mechanisms, rogue device detection.

EDR & Security Tools: SentinelOne, CrowdStrike, device control, vulnerability management, and desktop firewall policy tuning.

Programming & Automation: Python, Bash, PowerShell, SQL, automation scripts for IR, log parsing, reporting.

Service Management: Incident handling, SLA adherence, stakeholder communication, reporting, ticket workflows.

Soft Skills: Analytical Thinking, Clear Communication, Cross-Team Collaboration, Client-Facing Support.

EXPERIENCE

Huawei Technologies Co., Ltd (FESCO Adecco)

China

Senior Cybersecurity Analyst

Feb 2023 – Mar 2024

- Delivered **security architecture consulting** and conducted **comprehensive risk assessments** for enterprise clients across 7+ countries, aligning with **ISO 27001** and Huawei security compliance frameworks.
- Led **incident response** activities, including log correlation, forensic analysis, and malware triage, supporting containment and eradication in collaboration with SOC and IR teams.
- Performed **vulnerability assessment**, **penetration testing**, and **attack simulation** on Huawei Cloud and network infrastructure, identifying risks in IAM, API, and data protection layers.
- Investigated endpoint protection alerts using **SentinelOne Wireshark**, performing root-cause analysis, threat scoping, containment recommendations, and incident documentation.
- Maintained strong Tier 1 fundamentals and contributed to **SOC process improvements**, enhancing investigation quality and operational efficiency.
- Authored detailed **security testing reports**, technical presentations, and risk mitigation plans, strengthening client relationships and improving **security posture visibility**.

- Performed **white-box security assessments** through in-depth **source code review** (C, C++, Java, Shell), uncovering critical vulnerabilities including **SQL injection**, **RCE**, and **privilege escalation**.
- Developed **automated testing scripts** (Python, Bash) to streamline vulnerability scanning, log analysis, and compliance verification, increasing efficiency by over 40%.
- Executed **malware analysis** and **binary reverse engineering** using **IDA Pro**, **Ghidra**, and **Radare2**, strengthening detection of persistence, lateral movement, and code injection techniques.
- Supported **identity and access management (IAM)** testing by validating authentication flows, token management, and session handling security within enterprise applications.
- Collaborated with SOC and DFIR teams to correlate alerts, simulate attacks, and improve incident response playbooks for authentication abuse and insider threat detection.

PROJECTS

Malware Analysis Using Machine Learning | *CyberScienceLab with Professor Ali Dehghantanha*

- Developed **Python scripts** to automate the **bulk downloading of malware samples** from **VirusShare** and organized them by **APT groups**.
- Utilized **IDA Pro** to perform **opcode-level analysis** on the malware samples, gaining detailed insights into the **malicious code structure**.
- Engineered **Python automation** to **extract opcode sequences** in bulk from the analyzed malware, streamlining the data processing pipeline.
- Cleaned and structured the extracted **opcode data** into **CSV datasets**, forming the foundation for AI-based classification.
- Trained **KNN**, **Decision Trees**, and **SVM models** to classify **APT groups** based on opcode patterns, achieving **85% accuracy**—designed to **detect and defend against malware** in early stages of infection.

C2 Trojan Malware Development | *UofG with Professor Hassan Khan*

- Researched and implemented **phishing techniques** and **PDF-based malware embedding** to simulate **delivery via malicious email links**.
- Integrated **environment detection**, **string obfuscation**, and **AES encoding** techniques to **bypass or reduce EDR detection**.
- Developed **C2 (Command-and-Control)**, **ransomware (file encryption)**, and **keylogging** functionalities to simulate real-world **adversarial behavior**.
- Engineered **persistent foothold** via **scheduled background tasks**, mimicking how advanced malware maintains control.
- Aimed to **understand the mindset of malicious attackers** to better develop **defensive strategies** and enhance **cybersecurity posture**.

Privacy Information Attack and Protect Research Using AI | *UofG with Professor Rozita Dara*

- Spearheaded a research project on **privacy attacks and defense techniques in AI**, focusing on the impact of **data manipulation**, **model inversion**, and **membership inference** on user privacy.
- Engineered and executed **adversarial attack scenarios**, including **data poisoning**, **evasion attacks**, and **backdoor attacks**, to evaluate the vulnerability of AI models trained on **sensitive personal data**.
- Developed **Python-based simulation tools** for **model extraction** and **deepfake generation** to assess **privacy breach risks** and inform **mitigation strategies**.
- Designed and validated **robust defense mechanisms** to enhance **model robustness**, **fairness**, and **privacy protection**, leveraging **differential privacy** and **adversarial training** methodologies.

CTFs & WORKSHOPS

- **Crack The Code CTF**: Ranked in the **Top 10** in Splunk's *Boss of The SOC* CTF, demonstrating skills in threat detection, SOC operations, and log analysis.
- **ISA Vulnerability Assessment**: Performed security assessments using industry tools, identifying vulnerabilities and implementing best practices for system hardening.
- **TRU eSentire Threat Hunting CTF**: Conducted threat hunting exercises using **SIEM** and **YARA rules** to detect and analyze zero-day attacks and phishing threats.